

REPORT ASSIGNMENT 3

Task 1 : Model Architecture and Parameter Analysis

The following section details the dimensional properties of the model's tensors and parameters during execution.

1.1 Tensor Shapes

The fundamental input and output shapes of the model's forward pass are defined as follows:

- Input IDs Shape: torch.Size([1, 21])
- Attention Mask Shape: torch.Size([1, 21])
- Token Type IDs Shape: torch.Size([1, 21])
- Token Embedding Shape: torch.Size([1, 21, 512])
- Positional Embedding Shape: torch.Size([1, 21, 512])
- Segment Embedding Shape: torch.Size([1, 21, 512])
- Encoder Layer 1 Output Shape: torch.Size([1, 21, 512])
- Encoder Layer 2 Output Shape: torch.Size([1, 21, 512])
- Classifier Output Shape: torch.Size([1, 2])

1.2 Embedding Weights

- Token Embedding Weight: [30522, 512]
- Position Embedding Weight: [512, 512]
- Token Type Embedding Weight: [2, 512]

1.3 Parameter Layer Dimensions

Table 1: Shape of Each Parameter

Layer Component	Weight Shape	Bias Shape
Layer 0: Attention Query (q_linear)	[512, 512]	[512]
Layer 0: Attention Key (k_linear)	[512, 512]	[512]
Layer 0: Attention Value (v_linear)	[512, 512]	[512]
Layer 0: Attention Output (fc_out)	[512, 512]	[512]
Layer 0: Feed Forward 1 (fc1)	[2048, 512]	[2048]
Layer 0: Feed Forward 2 (fc2)	[512, 2048]	[512]
Layer 0: Normalization 1 (norm1)	[512]	[512]
Layer 0: Normalization 2 (norm2)	[512]	[512]
Layer 1: Attention Query (q_linear)	[512, 512]	[512]

Layer 1: Attention Key (k_linear)	[512, 512]	[512]
Layer 1: Attention Value (v_linear)	[512, 512]	[512]
Layer 1: Attention Output (fc_out)	[512, 512]	[512]
Layer 1: Feed Forward 1 (fc1)	[2048, 512]	[2048]
Layer 1: Feed Forward 2 (fc2)	[512, 2048]	[512]
Layer 1: Normalization 1 (norm1)	[512]	[512]
Layer 1: Normalization 2 (norm2)	[512]	[512]
Pooler	[512, 512]	[512]
Classifier	[2, 512]	[2]

Task 2: Data Preprocessing

2.1 Dataset Reduction and Distribution

The original dataset was substantial, containing 1,745,767 posts. This was downsampled to a working dataset of 5,000 posts. The sampled dataset is perfectly balanced, comprising 2,500 fake posts and 2,500 non-fake posts.

Table 2: database head samples

ID	Clean Title	2 way label
cvolg1	the way my binder rings open	1
4ofyrj	hating point a political cartoon satirising th...	0
clq0wm1	i was just hanging out in the spc kicking it w...	0
9bktsq	a cloud circling a cloud	1
6m0rnb	how friends saved a woman waiting for a heart ...	1

2.2 Train and Test Splits

The dataset was divided into training and testing subsets, maintaining class balance across both:

- Training Set: 4,000 samples (2,000 class '1', 2,000 class '0').
- Test Set: 1,000 samples (500 class '1', 500 class '0').

Task 3: Experimental Setup and Model Evaluation

3.1 Input Truncation Method

Truncation of the input data was applied at two distinct levels: dataset-level sampling and token-level truncation.

1. 1) Dataset-Level Truncation (Sampling-Based Reduction):
 - The dataset was initially filtered to retain only the relevant columns.
 - A fixed number of samples were selected from each class, which were then combined to establish a total dataset size of $n = 5000$.
2. 2) Shuffling and Final Preparation:
 - The dataset was shuffled to mitigate ordering bias.
 - Stratified sampling was utilized to split the dataset into training (80% $\rightarrow$ 4000 samples) and testing (20% $\rightarrow$ 1000 samples) sets.

3.2 Hyperparameters

output_dir: './results'
 num_train_epochs: 3
 per_device_train_batch_size: 16
 per_device_eval_batch_size: 16
 learning_rate: 2e-5
 warmup_steps: 100
 weight_decay: 0.01
 eval_strategy: 'epoch'
 save_strategy: 'epoch'
 logging_strategy: 'steps'
 logging_steps: 50

3.3 Base Model Architecture

The foundation of the architecture is a BertForSequenceClassification model, comprised of the following key sub-modules:

- Embeddings (BertEmbeddings): Incorporates Word Embeddings (30522, 768, padding_idx=0), Position Embeddings (512, 768), and Token Type Embeddings (2, 768), followed by LayerNorm and Dropout ($p=0.1$).
- Encoder (BertEncoder): Contains 12 BertLayer blocks (0-11). Each block consists of:
 - - BertAttention with Self-Attention (Query, Key, Value linear layers of 768x768) and a Dropout of 0.1.
 - - BertSelfOutput featuring a Dense layer (768x768), LayerNorm, and Dropout.
 - - BertIntermediate block mapping 768 features to 3072 using a GELU activation function.
 - - BertOutput block projecting back from 3072 to 768 with LayerNorm and Dropout.
- Pooler and Classifier: A BertPooler dense layer (768x768) with Tanh activation passes into a final Dropout layer and a Linear Classifier mapping the 768 features to 2 output classes.

3.4 Performance Metrics

Table 3: Training and Validation Progression

Epoch	Training Loss	Validation Loss	Validation Accuracy	Validation F1	Validation Precision	Validation Recall	Training Accuracy	Training F1	Training precision	Training Recall
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1	1.193273	0.895391	0.80500	0.819277	0.763385	0.884000	0.82375	0.8345	0.7863	0..889
2	0.796470	0.837949	0.806000	0.819030	0.76748	0.878000	0.89675	0.90112	0.86449	0.941
3	0.595006	0.870802	0.81700	0.82111	0.803059	0.840000	0.92575	0.92653	0.91679	0.9365

Table 4: Classification Report

Class	Precision	Recall	F1-Score	Support
0	0.83	0.79	0.81	500
1	0.80	0.84	0.82	500
accuracy			0.82	1000
Macro Avg	0.82	0.82	0.82	1000
Weighted Avg	0.82	0.82	0.82	1000

Confusion Matrix:

True Positives (Class 1): 397

True Negatives (Class 0): 420

False Positives (Type I Error): 103

False Negatives (Type II Error): 80

4. Task 4: Token Attribution using Integrated Gradients

Example 1:

- Text: jessica long meets biological parents in russia in heartwarming reunion
- Predicted class: 0
- Convergence delta: -4.607687950134277
- Top-5 Important Tokens: reunion (-0.1859), ##war (0.1373), jessica (-0.1275), meets (-0.1051), long (-0.0919)

Example 2:

- Text: michael jacksons ghost tells lionel richies exwife he accidentally killed himself hears court
- Predicted class: 0
- Convergence delta: 3.8079144954681396
- Top-5 Important Tokens: tells (-0.1424), accidentally (-0.1331), ex (-0.0888), ghost (-0.0767), michael(-0.0560)

Example 3:

- Text: my steak looks like the millennium falcon
- Predicted class: 0
- Convergence delta: 0.21279436349868774

- Top-5 Important Tokens: steak (-0.7237),millennium (-0.2463),falcon (-0.2016) ,the (0.0918) ,like (0.0657)

Example 4:

- Text: official photo of the hong kong protests from the chinese government
- Predicted class: 0
- Convergence delta: -0.5505692958831787
- Top-5 Important Tokens:photo(-0.5460),government(-0.3547) ,of(-0.3268) ,official(-0.2009) ,protests(-0.1844)

Example 5:

- Text: trump hails aretha franklin as pretty good for a negress
- Predicted class: 0
- Convergence delta:2.710731029510498
- Top-5 Important Tokens: ##tha(-0.2719), trump(-0.0763), hail(-0.0609), as(-0.0501), ##gre(-0.0426)

Example 6:

- Text: omanyte used water gun
- Predicted class: 0
- Convergence delta: -0.49277663230895996
- Top-5 Important Tokens: oman(-0.6241), gun(-0.4082), used(-0.3351), water(-0.1898), ##yte(-0.1064)